

Percent Increase, Percent Decrease, and Percent Change

When a quantity changes from an initial positive amount to another positive amount (for example, an employee's salary that is raised), you can compute the amount of change as a percent of the initial amount. This is called **percent change**. If a quantity increases from 600 to 750, then the base of the increase is the initial amount, 600, and the amount of the increase is $750 - 600$, or 150. The **percent increase** is found by dividing the amount of increase by the base, as follows.

$$\frac{\text{amount of increase}}{\text{base}} = \frac{750 - 600}{600} = \frac{150}{600} = \frac{25}{100} = 25\%$$

We say the percent increase is 25%. Sometimes this computation is written as follows.

$$\left(\frac{750 - 600}{600} \right) (100\%) = \left(\frac{150}{600} \right) (100\%) = 25\%$$

If a quantity doubles in size, then the percent increase is 100%. For example, if a quantity increases from 150 to 300, then the percent increase is calculated as follows.

$$\frac{\text{amount of increase}}{\text{base}} = \frac{300 - 150}{150} = \frac{150}{150} = \frac{25}{100} = 100\%$$

If a quantity decreases from 500 to 400, calculate the **percent decrease** as follows.

$$\frac{\text{amount of decrease}}{\text{base}} = \frac{500 - 400}{500} = \frac{100}{500} = \frac{20}{100} = 20\%$$

The quantity decreased by 20%.

When computing a percent increase, the base is the smaller number. When computing a percent decrease, the base is the larger number. In either case, the base is the initial number, before the change.

Example-15: An investment in a mutual fund increased by 12% in a single day. If the value of the investment before the increase was \$1,300, what was the value after the increase?

Solution: The percent increase is 12%. Therefore, the value of the increase is 12% of \$1,300, or, using the decimal equivalent, the increase is $(0.12)(\$1,300) = \156 . Thus, the value of the investment after the change is

$$\% \text{ change} = \frac{F.V. - I.V.}{I.V.} \times 100 \quad \$1,300 + \$156 = \$1,456$$

Because the final result is the sum of the initial investment (100% of \$1,300) and the increase (12% of \$1,300), the final result is $100\% + 12\% = 112\%$ of \$1,300. Thus, another way to get the final result is to multiply the value of the investment by the decimal equivalent of 112%, which is 1.12:

$$(\$1,300)(1.12) = \$1,456$$

A quantity may have several successive percent changes, where the base of each successive change is the result of the preceding percent change, as is the case in the following example.

Example-16: On September 1, 2013, the number of children enrolled in a certain preschool was 8% less than the number of children enrolled at the preschool on September 1, 2012. On September 1, 2014, the number of children enrolled in the preschool was 6% greater than the number of children enrolled in the preschool on September 1, 2013. By what percent did the number of students enrolled change from September 1, 2012 to September 1, 2014?

Solution: The initial base is the enrollment on September 1, 2012. The first percent change was the 8% decrease in the enrollment from September 1, 2012, to September 1, 2013. As a result of this decrease, the enrollment on September 1, 2013, was $(100 - 8)\%$, or 92%, of the enrollment on September 1, 2012. The decimal equivalent of 92% is 0.92.

So, if n represents the number of children enrolled on September 1, 2012, then the number of children enrolled on September 1, 2013, is equal to $0.92n$.

The new base is the enrollment on September 1, 2013, which is $0.92n$. The second percent change was the 6% increase in enrollment from September 1, 2013, to September 1, 2014. As a result of this increase, the enrollment on September 1, 2014, was $(100 + 6)\%$, or 106%, of the enrollment on September 1, 2013. The decimal equivalent of 106% is 1.06.

Thus, the number of children enrolled on September 1, 2014, was $(1.06)(0.92n)$, which is equal to $0.9752n$.

The percent equivalent of 0.9752 is 97.52%, which is 2.48% less than 100%. Thus, the percent change in the enrollment from September 1, 2012 to September 1, 2014, is a 2.48% decrease.

Exercise. Find the following.

(a) 40% of 15

Solution: $40\% \text{ of } 15 = \frac{40}{100} \times 15 = 6$

(b) 150% of 48

Solution: $15\% \text{ of } 485 = \frac{150}{100} \times 48 = 72$

(c) 0.6% of 800

Solution: $0.6\% \text{ of } 800 = \frac{6}{10} \times \frac{1}{100} 800 = 4.8$

(d) 15 is 30% of which number?

Solution: $30\% x = 15, \text{ or } 0.30 \times x = 15, x = \frac{15}{0.30} = 50$

(e) 11 is what percent of 55?

Solution: Here the whole is 55 is 11, so $\frac{\text{part}}{\text{whole}} = \frac{11}{55} = 0.2 = 20\%$

Unit 3A

USES AND ABUSES OF PERCENTAGE

Using and Understanding Mathematics: A Quantitative Reasoning Approach (6th Ed.) by Jeffrey Bennett, William Briggs

News reports frequently express quantitative information with percentages. Unfortunately, while percentages themselves are rather basic—they are just an alternative form of fractions—they are often used in very subtle ways. For example, consider the following quote that appeared in a front-page news article:

The rate of smoking among eighth graders was up 44 percent, to 10.4 percent. The percentages in this statement are used correctly, but the phrase “up 44%, to 10.4%” is not easy to interpret correctly. In this unit, we will investigate this statement (see Example 11), along with other subtle uses and abuses of percentages.

Three Ways of using Percentages

Consider the following statements from news reports:

- 1) A total of 13,000 newspaper employees, 2.6% of the newspaper work force, lost their jobs.
- 2) Citigroup stock fell 15% last week, to \$44.25.
- 3) The advanced battery lasts 125% longer than the standard one, but costs 200% more.

On close examination, each statement uses a percentage in a different way. The first uses a percentage to express a *fraction* of the total work force. The second uses a percentage to describe a *change* in stock price. The third uses percentages to *compare* the performance and costs of batteries.

Using Percentages as Fractions

Percent is just a fancy way of saying “divided by 100,” so $P\%$ simply means $P/100$. For example, 10.4% means $10.4/100$, or 0.104. Therefore, if 10.4% of eighth-graders smoke and there are 100,000 eighth-graders, then the number who smoke is 10.4% of 100,000, or $10.4\% \times 100,000 = 0.104 \times 100,000 = 10,400$. Notice that the word *of* told us to multiply. We’ve found that if 10.4% of 100,000 eighth-graders smoke, there are 10,400 smokers.

Brief Review Percentages

The words *per cent* mean “per 100” or “divided by 100.” We use the symbol % as shorthand for *per cent*. For example, we read 50% as “50 percent” and its meaning is

$$50\% = \frac{50}{100} = 0.5$$

More generally, for a number P ,

$$P\% = \frac{P}{100}$$

For example:

$$100\% = \frac{100}{100} = 1 \quad 200\% = \frac{200}{100} = 2$$

$$350\% = \frac{350}{100} = 3.5$$

Note that multiplying a number by 100% does not change its value, because 100% is just another way of writing 1. For example, multiplying 1.25 by 100% gives

$$1.25 \times 100\% = 125\%$$

That is, 125% is just another way of writing 1.25. These basic ideas lead to the following rules for converting between percentages and decimals or common fractions.

- To convert a percentage to a common fraction: Replace the % symbol with division by 100; simplify the fraction if necessary.

Example: $25\% = \frac{25}{100} = \frac{1}{4}$

- To convert a percentage to a decimal: Drop the % symbol and divide by 100 (equivalent to moving the decimal point two places to the left).

Example: $25\% = \frac{25}{100} = 0.25$

- To convert a decimal to a percentage: Multiply by 100 (equivalent to moving the decimal point two places to the right) and insert the % symbol.

Example: $0.43 = \frac{43}{100} = 43\%$

- To convert a common fraction to a percentage: First convert the common fraction to decimal form, using a calculator if necessary; then convert the decimal to a percentage.

Example: $\frac{1}{5} = 0.2 = 20\%$

► Now try Exercises 17–32.

150% = 1.5, Carol earns 1.5 times as much as William. For example, if Carol earns \$60,000 and William earns \$40,000, then Carol's salary is 150% of William's salary or 50% more than William's salary.

(Same as Q. 61–64)

Example-7: Sale!

A store is having a "25% off" sale. How does an item's sale price compare to its original price?

Solution The "25% off" means that an item's sale price is 25% *less than* its original price. The sale price is $100 - 25\% = 75\%$ of the original price. For example, if the original price is \$100, the sale price is \$75.

(Same as Q. 65–68)

Percentages of Percentages

Percentage changes or comparisons can be particularly confusing when the values themselves are percentages. Suppose a bank increases its mortgage interest rate from 3% to 4%. It's tempting to say that the interest rate increased by 1%, but this statement is ambiguous at best. The interest rate increased by 1 percentage point, but the relative change in the interest rate was

$$\text{Relative change} = \frac{\text{New value} - \text{Reference value}}{\text{Reference value}} \times 100\% = \frac{4\% - 3\%}{3\%} \times 100\% = 33\%$$

Brief Review**What Is a Ratio?**

Suppose we want to compare two quantities, such as the prices of a BMW that costs \$80,000 and a Honda that costs \$20,000. We can, of course, find the absolute or relative difference in the quantities. But another way to make the comparison is to compute the *ratio* of the two quantities. In this case, the two quantities are \$80,000 and \$20,000, so their ratio is

$$\frac{\$80,000}{\$20,000} = \frac{4}{1} = 4$$

Notice how the units of *dollars* canceled on the top and bottom of the fraction. This is a general characteristic of ratios: Because comparisons make sense only when we compare quantities with the same units, ratios always end up without units. Also note that we can state this result in several equivalent ways:

- The ratio of the BMW price to the Honda price is 4 to 1.
(The ratio can also be stated as 4, but "4 to 1" is more common.)

- The BMW price is 4 times the price of the Honda.
- The ratio of the Honda price to the BMW price is 1 to 4, which we can also state as $\frac{1}{4}$, 0.25, or 25%.

Example: Earth's average density is about 5.5 grams per cubic centimeter, and the average density of Saturn is about 0.7 gram per cubic centimeter. What is the ratio of their densities?

Solution: We divide Earth's density by Saturn's density:

$$\frac{\text{average density of Earth}}{\text{average density of Saturn}} = \frac{5.5 \text{ g/cm}^3}{0.7 \text{ g/cm}^3} \approx 8$$

The ratio of Earth's average density to Saturn's average density is about 8 to 1. Alternatively, we can say that Earth's density is about 8 times that of Saturn or that Saturn's density is about $\frac{1}{8}$ that of Earth. Again, notice how the units canceled, leaving the ratio without units.

► Now try Exercises 33–42.

In other words, you can say that the bank raised the interest rate by 33%, even though the actual rate increased by only 1 percentage point. More specifically, the absolute change in the interest rate was 1 percentage point, while the relative change in the interest rate was 33%.

Percentage Points versus %

When you see a change or difference expressed in percentage points, you can assume it is an absolute change or difference. If it is expressed with the % sign or the word percent, it should be a relative change or difference.

Example-8: Newspaper Readership Declines

The percentage of adults who report reading a daily print newspaper fell from about 78% in 1980 to 23% in 2013. Describe this change in newspaper readership.

Solution The drop in readership from 78% to 23% represents a decline of $78 - 23 = 55$ percentage points. This is the absolute change in the percentage of adults who read a daily newspaper.

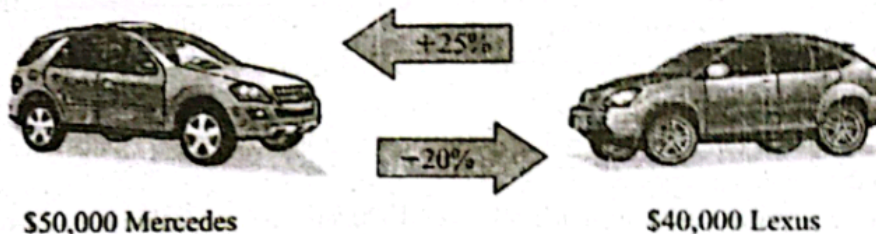
The relative change in readership is

$$\frac{\text{New value} - \text{Reference value}}{\text{Reference value}} \times 100\% = \frac{23\% - 78\%}{78\%} \times 100\% \approx -71\%$$

The negative sign indicates a decrease in readership. We say that readership dropped by about 71% in relative terms or by 55 percentage points in absolute terms.

(Same as Q. 69–72)

- 1) The Mercedes costs 25% more than the Lexus; or
- 2) The Lexus costs 20% less than the Mercedes.



Both statements are correct, yet they contain *different* percentages. That is why it is so important to keep careful track of the reference and compared values.

For those who like formulas, we can write the following rule: If quantity A is $P\%$ more than quantity B, then quantity B is $\frac{100P}{100+P}$ percent less than quantity A.

Example-5: Pay Comparison

Average pay for full-time wage earners varies from state to state. Recent data (data for 2009 released in 2013) show that New York ranked first in average pay, at about \$60,300 per person, and South Dakota ranked last at \$32,800 per person. Compare average pay in South Dakota to that in New York in both absolute and relative terms.

Solution Based on the wording of the statement, we use South Dakota pay as the compared value and New York pay as the reference value.

$$\begin{aligned}
 \text{Absolute difference} &= \text{Compared Value} - \text{Reference Value} = \$32,800 - \$60,300 = -\$27,500 \\
 \text{Relative difference} &= \frac{\text{Compared value} - \text{Reference value}}{\text{Reference value}} \times 100\% \\
 &= \frac{\$32,800 - \$60,300}{\$60,300} \times 100\% = -45.6\%
 \end{aligned}$$

In absolute terms, average pay in South Dakota was \$27,500 less than average pay in New York. In relative terms, average pay in South Dakota was about 46% less than average pay in New York. (You can verify that average pay in New York was about 84% more than average pay in South Dakota.) (Same as Q. 55–60)

Of versus More Than

Consider a population that triples in size from 200 to 600. There are two equivalent ways to state this change with percentages:

- 1) **Using more than:** The new population is 200% more than the original population. Here we are stating the *relative change* in the population:

$$\text{Relative Change} = \frac{\text{New value} - \text{Reference value}}{\text{Reference value}} \times 100\% = \frac{600 - 200}{200} \times 100\% = 200\%$$

- 2) **Using of:** The new population is 300% of the original population, which means it is three times the size of the original population. Here we are looking at the ratio of the new population to the original population:

$$\frac{\text{New Population}}{\text{Original Population}} = \frac{600}{200} = 3.00 = 300\%$$

Notice that the percentages in the *of* and *more than* statements are related by $300\% = 100\% + 200\%$. This leads to the following general relationships.

Of versus More Than (or Less Than)

- 1) If the new or compared value is $P\%$ more than the reference value, it is $(100 + P)\%$ of the reference value.
- 2) If the new or compared value is $P\%$ less than the reference value, it is $(100 - P)\%$ of the reference value.

Example-6: Salary Difference (BS.22)

Carol earns 50% more than William. How many times larger is her income than his?

Solution We use the rule that $P\%$ more than means $(100 + P)\%$ of. Because Carol's income is 50% more than William's, we set $P = 50$. Therefore, Carol's income is $(100 + 50)\% = 150\%$ of William's income. Because

Absolute change = new value – reference value = 7.1 billion – 2.7 billion = 4.4 billion

Relative change = $\frac{\text{new value} - \text{reference value}}{\text{reference value}} \times 100\% = \frac{7.1 \text{ billion} - 2.7 \text{ billion}}{2.7 \text{ billion}} \times 100\% = 163\%$

World population increased by 4.4 billion people, or by 163%, from 1953 to 2013. (Same as Q. 51–52)

Example-4: Depreciating a Computer

You bought a computer three years ago for \$1000. Today, it is worth only \$300. Describe the absolute and relative change in the computer's value.

Solution The reference value is the original price of \$1000, and the new value is its current worth of \$300. The absolute change in the computer's value is

Absolute change = new value – reference value = \$300 – \$1000 = – \$700

The negative sign tells us that the computer's current worth is \$700 *less than* the price you paid three years ago. The relative change is

Relative change = $\frac{\text{new value} - \text{reference value}}{\text{reference value}} \times 100\%$
 $= \frac{\$300 - \$1000}{\$1000} \times 100\% = -70\%$ (Same as Q. 53–54)

Again, the negative sign tells us that the computer is now worth 70% *less than* it was three years ago.

Using Percentages for Comparisons

Percentages are also commonly used to compare two numbers. Suppose we want to compare the price of a \$50,000 Mercedes to the price of a \$40,000 Lexus. The difference between the Mercedes and Lexus prices is

$\$50,000 - \$40,000 = \$10,000$

That is, the Mercedes costs \$10,000 *more than* the Lexus. We can also express this difference as a percentage of the Lexus price:

$\frac{\$10,000}{\$40,000} = 0.25 = 25\%$

In relative terms, the Mercedes costs 25% *more than* the Lexus.

Because we are *comparing to* the Lexus price, we say that the Lexus price is the **reference value**. Notice that the reference value follows the word *than*. The Mercedes price is the **compared value**. We can now define the absolute and relative *difference* between the quantities, much as we defined absolute and relative change. The absolute and relative differences are *positive* if the compared value is greater than the reference value and *negative* if the compared value is less than the reference value.

Absolute and Relative Difference

The **absolute difference** is the actual difference between the compared value and the reference value:

Absolute difference = Compared Value – Reference Value

The **relative difference** describes the size of the absolute difference in comparison to the reference value and can be expressed as a percentage:

Relative difference = $\frac{\text{Compared value} - \text{Reference value}}{\text{Reference value}} \times 100\%$

That's essentially all there is to comparisons, except for one subtlety: There's no particular reason why we chose the Lexus price as the reference value. We can just as easily use the Mercedes price as our reference value. In that case, the Lexus price is the compared value and the **absolute difference is**

Lexus price – Mercedes price = \$40,000 – \$50,000 = – \$10,000

The negative sign tells us that the Lexus costs \$10,000 *less than* the Mercedes.

The relative difference is: $\frac{\text{Lexus Price} - \text{Mercedes Price}}{\text{Mercedes Price}} \times 100\% = \frac{-\$10,000}{\$50,000} \times 100\% = -20\%$

The negative sign indicates that the Lexus costs 20% *less than* the Mercedes.

We now have two ways to express the relative difference:

Example-1: Presidential Survey (BS.22)

An opinion poll finds that 64% of 1069 people surveyed said that the President is doing a good job. How many said the President is doing a good job?

Solution

$$\frac{64}{100} \times 1069 = 684.16 \approx 684$$

Because of indicates multiplication, 64% of the 1069 respondents is $64\% \times 1069 = 0.64 \times 1069 = 684.16 \approx 684$. About 684 people said the President is doing a good job. We rounded the answer to 684 to obtain a whole number of people. (Same as Q. 43–48)

Using Percentages to describe Change

Percentages are often used to describe how a quantity changes with time. As an example, suppose the population of a town was 10,000 in the 1980 census and 15,000 in the 2010 census. We can express the change in population in two basic ways:

- 1) Because the population rose by 5000 people (from 10,000 to 15,000), we say that the **absolute change** in the population was 5000 people.
- 2) Because the increase of 5000 people was 50% of the starting population of 10,000, we say that the **relative change** in the population was 50% or 0.5.

In general, calculating an absolute or relative change always involves two numbers: a starting number, or **reference value**, and a **new value**. Once we identify these two values, we can calculate the absolute and relative change with the formulas in the following box. Note that the absolute and relative changes are *positive* if the new value is greater than the reference value and *negative* if the new value is less than the reference value.

Absolute and Relative Change

The **absolute change** describes the actual increase or decrease from a reference value to a new value:

$$\text{Absolute Change} = \text{New Value} - \text{Reference Value}$$

The **relative change** is the size of the absolute change in comparison to the reference value and can be expressed as a percentage:

$$\text{Relative change} = \frac{\text{New Value} - \text{Reference Value}}{\text{Reference Value}} \times 100\%$$

As the next example shows, the relative change formula leads to the following important rules:

- 1) When a quantity *doubles* in value, its relative change is 100%.
- 2) When a quantity *triples* in value, its relative change is 200%.
- 3) When a quantity *quadruples* in value, its relative change is 300%.
- 4) When a quantity is *halved* in value, its relative change is -50%. And so on.

Example-2: Stock Price Rise

During a 6-month period, Xerox stock doubled in price from \$7 to \$14. What were the absolute and relative changes in the stock price?

Solution The reference value is the starting stock price of \$7, and the new value is the later stock price of \$14. The absolute change is the difference between the new and starting stock prices:

$$\text{Absolute change} = \text{New Value} - \text{Reference Value}$$

$$\text{Absolute change} = \$14 - \$7 = \$7$$

$$\text{Relative change} = \frac{\text{new value} - \text{reference value}}{\text{reference value}} \times 100\% = \frac{\$14 - \$7}{\$7} \times 100\% = 100\%$$

That is, the doubling of the stock price means a relative increase in value of 100%. (Same as Q. 49–50)

Example-3: World Population Growth

Estimated world population was 2.7 billion in 1953 and 7.1 billion in 2013. Describe the absolute and relative change in world population over this 60-year period.

Solution The reference value is the 1953 population of 2.7 billion, and the new value is the 2013 population of 7.1 billion.

- 47) 1189 people per square mile in New Jersey and a national average of 88.1 Americans per square mile in 2012.

Solution: 1189 is 1350% of 88.1 because $\frac{1189}{88.1} = 13.50$

- 48) An estimated 10,000 U.S. nuclear weapons out of an estimated 27,000 nuclear weapons worldwide (as of 2010)

Solution: $\frac{10,000}{27,000} = 0.3703 \Rightarrow 0.3703 \times 100 = 37.0 \approx 37$.

- 49) ✓ **Salary Comparisons.** Clint's salary increased from \$20,000 to \$28,000 over a three-year period. Helen's salary increased from \$25,000 to \$35,000 over the same period. Whose salary increased more in absolute terms? In relative terms? Explain.

Solution: Clint's Absolute change = \$28,000 - \$20,000 = \$8,000 , Relative change = $\frac{\$8,000}{\$20,000} \times 100 = 40\%$

Helen's Absolute change = \$35,000 - \$25,000 = \$10,000 , Relative change = $\frac{10000}{25000} \times 100 = 40\%$

Thus Helen's salary grew more in absolute terms, but the relative change was the same. This happened because we are comparing Helen's larger absolute change to her also larger initial salary.

- 50) **Population Comparison.** Between the 2010 U.S. census and 2011 (one year later), the official population of Denver increased from 600,158 to 619,968, During the same period the population of San Antonio increased from 1,327,407 to 1,359,758. Which city had the greater absolute change in population? Which city had the greater relative change in population?

Solution: Denver = 619,968 - 600,158 = 19,810, San Antonio = 1,359,758 - 1,327,407 = 32,351

Relative change = $\frac{(619,968 - 600,158)}{600,158} = 0.033 \approx 3.3\%$

Relative change = $\frac{(1359758 - 1327407)}{1327407} = 0.024 \approx 2.4\%$

- 51-54) **Percentage Change.** Find the absolute change and the percentage change in the following cases.

- 51) The average sale price of a house in the United States decreased from \$301,000 in February 2008 to \$152,000 in February 2013.

Solution: Absolute change = \$152,000 - \$301,000 = -\$149,000

Relative change = $\frac{(\$152,000 - \$301,000)}{\$301,000} \times 100 = -49.5\%$

- 52) The congressional delegation of California increased from 30 in 1950 to 53 in 2014.

Solution: Absolute change = 53 - 30 = 23 $\Rightarrow$ Relative change = $23/30 = 0.7666 = 76.170 \approx 76$

- 53) The number of daily newspapers in the United States was 2226 in 1900 and 1382 in 2011.

Solution: The absolute change is 1382 papers - 2226 papers = -844 papers.

The relative change is $\frac{1382 - 2226}{2226} = -0.398 = -37.9\%$.

- 54) Total revenue from music sales decreased from \$14.6 billion to \$5.5 billion between 2000 and 2014.

Solution: Absolute change = 5.5 billion - \$14.6 billion = - \$ 9.1 billion

Relative change = $\frac{(\$5.5 - \$14.6)}{\$14.6} \times 100\% = -62.33\%$

- 55-60) **Percentage Comparisons.** Complete the following sentences.

(b) Ratio of B to A means B divided by A $B/A = 30/10 = 3$

(c) A is _____ percent of B $= 0.33 \dots \times 100\% = 33\%$

35) A = 150 and B = 400 **Solution:** 150 to 400 = $3/8$, 400 to 150 = $8/3$, 150 is 37.5% of 400

36) A = 450 and B = 400

Solution: (a) Ratio of A to B means A by B $A/B = 450/400 = 9/8 = 1.125$

(b) Ratio of B to A means B divided by A $B/A = 400/450 = 8/9 = 0.88\dots$

(c) A is _____ percent of B $= 1.125 \times 100\% = 112.5\%$

37) A = 52,252 is the number of deaths due to AIDS in the United States in 1995, and B = 17,774 is the number of deaths due to AIDS in the United States in 2009 (most recent data available).

Solution: 52,252 to 17,774 = 2.94, 17,774 to 52,252 = 0.34, 52,252 is 294% of 17,774.

38) A = 40,229,000 is the number of Americans over age 65 in 2010, and B = 88,458,000 is the projected number of Americans over age 65 in 2050.

Solution: $\frac{A}{B} = \frac{40229000}{88458000} = 0.45$, $\frac{B}{A} = \frac{88458000}{40229000} = 2.199$ A is _____ percent of B $= 0.455 \times 100.1 = 45.51$.

39) A = 1.5 million is the 2012 population of Philadelphia, and B = 2.1 million is the 2012 population of Houston.

Solution: 1.5 million to 2.1 million = 0.71, 2.1 million to 1.5 million = 1.4, 1.5 million is 71% of 2.1 million.

40) A = 472 is the average mathematics SAT score in Maine, and B = 523 is the average mathematics SAT score in Vermont.

Solution: $\frac{A}{B} = \frac{472}{523} = 0.902$, $\frac{B}{A} = \frac{523}{472} = 1.108$ A is _____ percent of B $= 0.902 \times 100.1 = 90.21$

41) A = 1.82 million is the 2012 daily circulation of USA Today (second highest in the nation), and B = 2.12 million is the 2012 daily circulation of the Wall Street Journal (highest in the nation).

Solution: 1.82 million to 2.12 million = 0.86, 2.12 million to 1.82 million = 1.16, 1.82 million is 86% of 2.12 million.

42) A = 87% is the 2011 high school graduation rate in Wisconsin (first in the nation), and B = 62% is the high school graduation rate in Nevada (last in the nation).

Solution: A = **Solution:** $1 = 0.87$, $\Rightarrow A = 0.87$, $B = \frac{62}{100}$, $\Rightarrow 0.62 \Rightarrow B = 0.62$
 $\frac{A}{B} = \frac{0.87}{0.62} \times \frac{100}{100} = 1.403$, $\frac{B}{A} = \frac{0.62}{0.87} \times \frac{100}{100} = 0.713$, A is _____ percent of B $= 1.403 \times 100.1 = 140.31$

43-48) **Percentages as Fractions.** In the following statements, express the first number as a percentage of the second number.

43) 28 pounds of recyclable trash in a barrel of 52 pounds of trash

Solution 28 is 53.8% of 52 because $28/52 = 0.538$

44) 12.0 million metric tons of beef produced annually in the United States out of 65.1 million metric tons of beef produced annually worldwide.

Solution: A = 12.0 million, B = 65.1 million

A is the percentage of B $\frac{A}{B} = \frac{12.0 \text{ million}}{65.1 \text{ million}} \times 100.1 = 18.46$

45) The full-time year-round median salary for U.S. men in 2010 was \$42,800, and the full-time year-round median salary for U.S. women in 2010 was \$34,700 (most recent data).

Solution: $\frac{\$42800}{\$34700} = 1.233$ $\Rightarrow 1.233 \times 100 = 123.3 \approx 123\%$

46) 63 million Americans under age 15 out of 314 million Americans in 2011.

Solution: $\frac{63000000}{314000000} = 0.200$ $\Rightarrow 0.200 \times 100 = 20.06 \approx 20$.

Problems are more for practice

- 4) Explain the difference between the key words *of* and *more than* when dealing with percentages. How are their meanings related?
- 5) Explain the difference between the terms *percent (%)* and *percentage points*. Give an example of how they can differ for the same situation.
- 6) Give an example to explain why, in general, it is not legitimate to average percentages.

Does it make Sense

Decide whether each of the following statements makes sense (or is clearly true) or does not make sense (or is clearly false). Explain your reasoning.

- 7) In many European countries, the percentage change in population has been negative in recent decades. Make Sense
- 8) The price of tuition has tripled since my parents went to school—that's a 200% increase in price! Make Sense
- 9) My older child weighs 25% more than my younger child. Make Sense
- 10) I've decreased my caloric intake by 125% which has helped me lose weight. Does not Make Sense
- 11) If you earn 20% more than I do, then I must earn 20% less than you do. Does not Make Sense
- 12) If they raise taxes by 10% every year, in a decade we'll be paying everything we earn to taxes.
- 13) We found that these rare cancers were 700% more common in children living near the toxic landfill than in the general population. Make Sense
- 14) I have a 60% average on our assignments going into the final exam, but I still hope to raise my course average to 70% by getting an 80% on the final. (The final is worth 25% of the final grade.) Make Sense
- 15) The rate of return on our fund increased by 50%, to 15%. Make Sense
- 16) My bank increased the interest rate on my savings account 100%, from 2% to 4%.

Basic Skills & Concept

17–32: **Fractions, Decimals, Percentages.** Express the following numbers in three forms: as a reduced fraction, as a decimal, and as a percentage. (Refer to the Brief Review)

- | | | | |
|-----------|------------------------------|----------------------------|---|
| 17) $2/5$ | Fraction = $4/10 = 2/5$ | Decimal = 0.4 | Percentage = 40% |
| 18) 30% | Fraction = $3/10 = 3/10$ | Decimal = 0.3 | Percentage = 30% |
| 19) 0.20 | Fraction = $20/100 = 1/5$ | Decimal = 0.20 | Percentage = 20% |
| 20) 0.85 | Fraction = $85/100$ | Decimal = 0.85 | Percentage = 85% |
| 21) 150% | Fraction = $150/100 = 3/2$ | Decimal = 1.5 | Percentage = 150% |
| 22) $2/3$ | Fraction = $2/3$ | Decimal = 0.66... | Percentage = 66% |
| 23) $4/9$ | Fraction = $4/9$ | Decimal = 0.444... | Percentage = 44.44% |
| 24) 1.25 | Fraction = $1.25/100 = 5/4$ | Decimal = 1.25 | Percentage = $1.25 \times 100 = 125.1\%$ |
| 25) $5/8$ | Fraction = $5/8$ | Decimal = $5/8 = 0.625$ | Percentage = $0.625 \times 100 = 62.51\%$ |
| 26) 44% | Fraction = $44/100 = 11/25$ | Decimal = $11/25 = 0.44$ | Percentage = 44.1% |
| 27) 69% | Fraction = $69/100$ | Decimal = $69/100 = 0.69$ | Percentage = 69.1% |
| 28) 4.25 | Fraction = $4.25/100 = 17/4$ | Decimal = 4.25 | Percentage = $4.25 \times 100 = 425.1\%$ |
| 29) $7/5$ | Fraction = $7/5$ | Decimal = $7/5 = 1.4$ | Percentage = $1.4 \times 100 = 140\%$ |
| 30) 121% | Fraction = $121/100$ | Decimal = $121/100 = 1.21$ | Percentage = 121% |
| 31) $4/3$ | Fraction = $4/3$ | Decimal = $4/3 = 1.33$ | Percentage = 133% |
| 32) 0.6 | Fraction = $0.60/100$ | Decimal = 0.60 | Percentage = $0.60 \times 100 = 60\%$ |

33–42) **Review of Ratios.** Compare the following pairs of numbers A and B in three ways:

- a) Find the ratio of A to B.
- b) Find the ratio of B to A.
- c) Complete the sentence: A is _____ percent of B. (Refer to the Brief Review)

33) A = 8 and B = 4

Solution

- (a) Ratio of A to B means A by B $A \div B$ $A/B = 8/4 = 2$
- (b) Ratio of B to A means B divided by A $B \div A$ $B/A = 4/8 = 1/2$ or 0.5
- (c) A is 200% of B $A/B = 2$ for percentage

34) A = 10 and B = 30

Solution:

- (a) Ratio of A to B means A by B $A/B = 10/30$, $1/3$ or 0.33

Example-9: Care in Wording

Assume that 40% of the registered voters in Carson City are Republicans. Read the following questions carefully, and give the most appropriate answers.

- The percentage of voters registered as Republicans is 25% higher in Freetown than in Carson City. What percentage of the registered voters in Freetown are Republicans?
- The percentage of voters registered as Republicans is 25 percentage points higher in Freetown than in Carson City. What percentage of the registered voters in Freetown are Republicans?

Solution

- We are told that the percentage of registered Republicans in Carson City is 40% and that the percentage in Freetown is 25% higher. We interpret the 25% as a relative difference. Because 25% of 40% is 10% ($0.25 \times 0.40 = 0.10$), we add this value to the Carson City percentage to find that the percentage of registered Republicans in Freetown is $40\% + 10\% = 50\%$.
- We interpret 25 percentage points as an absolute difference, so we add this value to the percentage of Republicans in Carson City. The percentage of registered Republicans in Freetown is $40\% + 25\% = 65\%$.

(Same as Q. 73–74)

Solving Percentage Problems

Consider this statement: Retail prices are 25% more than wholesale prices.

If you knew a wholesale price, how would you calculate the retail price? One way to begin is by translating the more than statement into an of statement. The statement becomes

Retail prices are $(100 + 25)\% = 125\%$ of wholesale prices.

Replacing of with multiplication, we write this statement mathematically as

Retail price = $125\% \times$ wholesale price

With this equation, we can find a retail price from a wholesale price. For example, if the wholesale price is \$10, the retail price is $125\% \times \$10 = 1.25 \times \$10 = \$12.50$. We can rearrange the same equation to find a wholesale price from a retail price. To do so, we first divide both sides of the equation by 125%:

$$\frac{\text{Retail Price}}{125\%} = \frac{125\% \times \text{Wholesale price}}{125\%}$$

We then interchange the left and right sides to write the result:

$$\text{Wholesale price} = \frac{\text{Retail price}}{125\%}$$

For example, if the retail price is \$15, the wholesale price is $\$15/1.25 = \12 . We can now generalize our results.

Example-10: Tax Calculation

- You purchase a shirt with a labeled (pre-tax) price of \$17. The local sales tax rate is 5%. What is your final cost (including tax)?
- Your receipt shows that you paid \$19.26 for a Blu-ray disc, tax included. The local sales tax rate is 7%. What was the labeled (pre-tax) price of the disc? (BS.22, ADS.22)

Solution

- The final cost of an item is its labeled price plus the sales tax. For a tax rate of $P\% = 5\%$, the final cost is 5% more than the labeled price. Changing from a more than statement to an of statement, we have

$$\text{Final cost} = (100 + 5)\% \times \text{labeled price} = 105\% \times \text{labeled price}$$

When we substitute the given labeled price of \$17, the final cost is

$$\text{Final cost} = 105\% \times \$17 = 1.05 \times \$17 = \$17.85$$

The final cost for the shirt is \$17.85.

- This time the tax rate is $P\% = 7\%$, so the relation between labeled price and final cost is

$$\text{Final cost} = (100 + 7)\% \times \text{labeled price} = 107\% \times \text{labeled price}$$

We are given the final cost of \$19.26. We find the labeled price by dividing both sides by $107\% = 1.07$:

$$\text{Labeled price} = \frac{\text{Final cost}}{107\%} = \frac{\$19.26}{107\%} = \frac{\$19.26}{1.07} = \$18.00$$

- 55) The gestation period of humans (266 days) is _____ percent longer than the gestation period of grizzly bears (220 days).

Solution: Absolute difference = $266 - 220 = 46$
 Relative difference = $\frac{266 - 220}{220} \times 100 = 20.9\%$

Which means the gestation period of humans (266 days) is 20.9 percent longer than the gestation period of grizzly bears (220 days).

- 56) The 2011 life expectancy in Japan (82.7 years) is _____ percent greater than the 2011 world average life expectancy (67.9 years).

Solution: Absolute difference = $82.7 - 67.9 = 14.8$
 Relative difference = $\frac{82.7 - 67.9}{67.9} \times 100 = 21.7\%$

- 57) The main span of the Tacoma Narrows bridge (2800 feet) is _____ percent shorter than the main span of the Golden Gate bridge (4200 feet).

Solution: Absolute difference = $2800 - 4200 = -1400$
 Relative difference = $\frac{2800 - 4200}{4200} \times 100 = -33.3\%$

Which means the main span of the Tacoma Narrows bridge (2800 feet) is 33.3 percent shorter than the main span of the Golden Gate bridge (4200 feet).

- 58) The median age at first marriage of U.S. women (26.1 years) is _____ percent less than the median age at first marriage of U.S. men (28.2 years).

Solution: Absolute difference = $26.1 - 28.2 = -2.1$
 Relative difference = $\frac{26.1 - 28.2}{28.2} \times 100 = -7.44\%$

- 59) The number of deaths due to poisoning in the United States in 2009 (39,000) is _____ percent greater than the number of deaths due to falls (26,100) (most recent data).

Solution: Absolute difference = $39,000 - 26,100 = 12,900$
 Relative difference = $\frac{39,000 - 26,100}{26,100} \times 100 = 49.4\%$

- 60) The amount of wheat produced by China in 2009 (115 million metric tons) was _____ percent greater than the amount of wheat produced by India in 2007 (81 million metric tons).

Solution: Absolute difference = $115 - 81 = 34$
 Relative difference = $\frac{115 - 81}{81} \times 100 = 41.9\%$

Will's height is 122% of Wanda's height because 22% more than expresses a relative difference in heights, and this is equivalent to 100% + 22% of the reference height (i.e. Wanda's height).

- 61-64) **Of versus More Than.** Fill in the blanks in the following statements.

- 61) Will is 22% taller than Wanda, so Will's height is _____ % of Wanda's height.

Solution: $(100+P)\%$ $P = 22\%$ $\Rightarrow$ $(100 + 22) \% = 122\%$

Will's height is 122% of Wanda's height because 22% more than expresses a relative difference in heights, and this is equivalent to 100% + 22% of the reference height (i.e. Wanda's height).

- 62) The area of Norway is 24% more than the area of Colorado, so Norway's area is _____ % of Colorado's area.

Solution: $(100+P)\%$ $P = 24\%$ $\Rightarrow$ $(100 + 24) \% = 124\%$

- 63) The population of Virginia is 18% less than the population of Georgia, so Virginia's population is _____ % of Georgia's population.

Solution: $(100 - P)\%$ $P = 18\%$ $\Rightarrow (100 - 18)\% = 82\%$

Virginia's population is 82% of Georgia's population because 18% less than expresses a relative difference in population, and this is equivalent to $100\% - 18\%$ of the reference population (i.e., Georgia's population).

- 64) The net worth of Zelda is 6.4% less than the net worth of Alicia, so Zelda's net worth is _____ % of Alicia's net worth.

Solution: $(100 - P)\%$ $P = 6.4\%$ $\Rightarrow (100 - 6.4)\% = 93.6\%$

- 65-68) **Prices and Sales.** Fill in the blanks in the following statements.

- 65) The wholesale price of a TV is 40% less than the retail price. Therefore, the wholesale price is _____ times the retail price.

Solution: $(100 - P)\%$ $P = 40\%$ $\Rightarrow (100 - 40)\% = 60\% = 0.6$

Since the wholesale price is 40% less than the retail price, it is 60% of the retail price, which means it is 0.6 times the retail price.

- 66) A store is having a 50% off sale. Therefore, the original price of an item is _____ times as much as the sale price.

Solution: $(100 - P)\%$ $P = 50\%$ $\Rightarrow (100 - 50)\% = 50\%$

- 67) The retail cost of a TV is 30% more than its wholesale cost. Therefore, the retail cost is _____ times the wholesale cost.

Solution: $(100 + P)\%$ $P = 30\%$ $\Rightarrow (100 + 30)\% = 130\% = 1.3$

Since the retail cost is 30% more than the wholesale cost, it is 130% of the wholesale cost, which means it is 1.3 times the wholesale cost.

- 68) A store is having a 40% off sale. The sale price for an item with a regular cost of \$80 is 48. *Price of 80% item = 64 / 100 = 48*

Solution: $(100 - P)\%$ $P = 40\%$ $\Rightarrow (100 - 40)\% = 60\%$

- 69-72) **Percentages of Percentages.** Describe each of the following changes in two ways: as an absolute change in terms of percentage points and as a relative change in terms of a percentage.

- 69) The annual interest rate for Jack's savings account increased from 2.3% to 2.8%.

Solution: Absolute change = $2.8 - 2.3 = 0.5$, Relative change = $\frac{2.8 - 2.3}{2.3} \times 100 = 21.7\%$

- 70) The percentage of Republicans in the House of Representatives increased from 40.9% in 2010 to 55.6% in 2012.

Solution: Absolute change = $55.6 - 40.9 = 14.7$, Relative change = $\frac{55.6 - 40.9}{40.9} \times 100 = 35.94\%$

- 71) The percentage of Americans accessing the Internet increased from 67% in 2000 to 83% in 2012.

Solution: Absolute change = $83 - 67 = 16$, Relative change = $\frac{83 - 67}{67} \times 100 = 23.9\%$

- 72) The percentage of fifth-graders who scored at or above the national average on the reading test rose to 44 percent from 25 percent.

Solution: Absolute change = $44 - 25 = 19$, Relative change = $\frac{44 - 25}{25} \times 100 = 76\%$

- 73) **Care in Wording.** Assume that 30% of city employees in Carson City ride the bus to work. Consider the following two statements:

- The percentage of city employees who ride the bus to work is 10% higher in Freetown than in Carson City.
- The percentage of city employees who ride the bus to work is 10 percentage points higher in Freetown than in Carson City.

Carson City: For each case, state the percentage of city employees in Freetown who ride the bus to work. Briefly explain why the two statements have different meanings.

Solution: In the first case, 33% of city employees in Freetown ride the bus to work because 10% more than 30% is 33%. In the second case, 40% of the city employees in Freetown ride the bus, because 40% is 10 percentage points more than 30%.

74) **Ambiguous News.** The average annual precipitation on Mt. Washington, New Hampshire, is 90 inches. During one particularly wet year, different news reports carried the following statements.

- The precipitation this year is 200% of normal.
- The precipitation this year is 200% above normal.

What does each of these statements imply about the precipitation during this year? Do the two statements have the same meaning? Explain.

Solution: 100% of normal would mean $100\% \times 90 = 90$ inches
 Normal 200% of normal would be $200\% \times 90 = 180$ inches
 Double normal 100% above normal would be $90 + 180 = 270$ inches

75-78) **Solving Percentage Problems.** Solve the following percentage problems.

75) You purchase a bicycle with a retail (pre-tax) price of \$760. The local sales tax rate is 7.6%. What is the final cost?

Solution: $(100 + P), P = 7.6\%$, $\text{retail price} = 760\$$
 $\text{tax amount} = \frac{7.6 \times 76}{10} = 57.76\$$
 $\text{Final cost} = 760\$ + 57.76\$ = 817.76\$$
 Final cost = $(100 + 7.6)\% \times \text{Labeled price} = 107.6\% \times \text{labeled price} = 1.076 \times 760\$ = 817.76\$$

76) The final cost of your new shoes is \$107.69. The local sales tax rate is 6.2%. What was the retail (pre-tax) price?

Solution: $(100 + P), P = 6.2\%$ Price = $(100 + 6.2)\% = 106.2\%$
 Lebal cost = $\$107.69 / 106.2\% = \101.4

77) Between 2000 and 2010, the percentage of U.S. households with cordless phones increased by 13.7% to 91%. What percentage of households had cordless phones in 2000?

Solution: $\frac{91\%}{1.137} = 80\%$ of households had cordless phones in 2000.

Price 100
 Final cost 106.2
 107.69

78) Between 2000 and 2010, the percentage of fatal automobile accidents due to speeding decreased by 34% to 16%. What percentage of fatal automobile accidents were due to speeding in 2000?

Solution: N.V = 16.1, P.V = 34.1, Decrease P% = $(100 - P).1 = (100 - 34).1 = 66.1$
 $P.R = \frac{N.R}{66.1} = \frac{16.1}{66.1} = 0.2424$, For 1 = $0.2424 \times \frac{100}{100} = 24.24.1$

79-82) **Shifting Reference value.** State whether the following statements are true or false, and explain why. If a statement is false, state the true change.

79) If the national economy shrank at an annual rate of 4% per year for three consecutive years, then the economy shrank by 12% over the three-year period.

Solution: False. If the national economy began at E, and it shrinks at 4% annually, it would be at $E \times 0.96 \times 0.96 \times 0.96 = 0.885E$, which is 11.5% less than the original value.

80) You receive a pay raise of 5%, then receive a pay cut of 5%. After the two changes in pay, your salary is unchanged.

81) If the profits in your consulting business increase by 4% one year and decrease by 2% the following year, your profits are up by 2% over two years.

Solution: False. Assign an arbitrary number to your profits before the increases, such as \$100. If your profits increase by 4% in the first year, they will be \$104. If they decrease by 2% in the next year, they will be $0.98 \times \$104 = \101.92 . This is an increase of 1.92%, not 2%.

82) A high school reports that its students' SAT scores were down by 20% for one year. The next year, however, SAT scores rose by 30%. The high school principal announces, "Overall, test scores have improved by 10% over the past two years."

83-88) **Is It Possible?** Determine whether the following claims could be true. Explain your answer.

83) By turning off her lights and closing her windows at night, Maria saved 120% on her monthly energy bill.

Solution: Not possible. A bill can decrease at most by 100%. Once you've deducted 100% from a bill, there's nothing more to pay.

84) Scott is 140% taller than his son.

85) Restaurant prices have increased 100% in the last 20 years.

Solution: Possible. An increase of 100% means restaurant prices are twice what they used to be.

86) Through hard training, Renee improved her 10-kilometer running time by 100%.

87) Your computer is 500% faster than mine.

MODULE 4**FINANCIAL ISSUES****HEC Syllabus 2022**

Financial issues: Stock exchange and economy, Money management (profit, loss, discount, zakat, simple interest, compound interest and taxation), Money management in practical life scenarios like investments and federal budget

Unit-M4.I

MONEY MANAGEMENT

(Profit, Loss, Discount, Taxation, and Other Scenarios)

Section 1.7**Percent**

The term **percent** means *per hundred*, or *hundredths*. Percents are ratios that are often used to represent *parts of a whole*, where the whole is considered as having 100 parts. Percents can be converted to fraction or decimal equivalents. Here are three examples of percents.

Example-1: 1 percent means 1 part out of 100 parts. The fraction equivalent of 1 percent is $\frac{1}{100}$, and the decimal equivalent is 0.01.

Example-2: 32 percent means 32 parts out of 100 parts. The fraction equivalent of 32 percent is $\frac{32}{100}$, and the decimal equivalent is 0.32.

Example-3: 50 percent means 50 parts out of 100 parts. The fraction equivalent of 50 percent is $\frac{50}{100}$, and the decimal equivalent is 0.50.

Note that in the fraction equivalent, the part is the numerator of the fraction and the whole is the denominator. Percents are often written using the percent symbol, %, instead of the word "percent." Here are five examples of percents written using the % symbol, along with their fraction and decimal equivalents.

Example-4: $100\% = \frac{100}{100} = 1$

Example- 5: $12\% = \frac{12}{100} = 0.12$

Example: 6: $8\% = \frac{8}{100} = 0.08$

Example-7: $10\% = \frac{10}{100} = 0.1$

Example-8: $0.3\% = \frac{0.3}{100} = 0.003$

Be careful not to confuse 0.01 with 0.01%. The percent symbol matters. For example,

$0.01 = 1\%$ but $0.01\% = \frac{0.01}{100} = 0.0001$.

To compute a percent, given the part and the whole, first divide the part by the whole to get the decimal equivalent, then multiply the result by 100. The percent is that number followed by the word "percent" or the % symbol.

Example-9: If the whole is 20 and the part is 13, you can find the percent as follows.

$$\frac{\text{part}}{\text{whole}} = \frac{13}{20} = 0.65 = 65\%$$

Example-10: What percent of 150 is 12.9? (BS.22)**Solution:** Here, the whole is 150 and the part is 12.9, so

$$\frac{\text{part}}{\text{whole}} = \frac{12.9}{150} = 0.86 = 8.6\%$$

To find the part that is a certain percent of a whole, you can either multiply the whole by the decimal equivalent of the percent or set up a proportion to find the part.

Example-11: To find 30% of 350, you can multiply 350 by the decimal equivalent of 30%, or 0.3, as follows.

$$(350)(0.3) = 105$$

Alternatively, to use a proportion to find 30% of 350, you need to find the number of parts of 350 that yields the same ratio as 30 parts out of 100 parts. You want a number x that satisfies the proportion

$$\frac{\text{part}}{\text{whole}} = \frac{30}{100} \text{ or } \frac{x}{350} = \frac{30}{100}$$

Solving for x yields $x = \frac{(30)(350)}{100} = 105$, so 30% of 350 is 105.

Given the percent and the part, you can calculate the whole. To do this, either you can use the decimal equivalent of the percent or you can set up a proportion and solve it.

Example-12: 15 is 60% of what number?**Solution:** Use the decimal equivalent of 60%. Because 60% of some number z is 15, multiply z by the decimal equivalent of 60%, or 0.6.

$$0.6z = 15$$

Now solve for z by dividing both sides of the equation by 0.6 as follows.

$$z = \frac{15}{0.6} = 25$$

Using a proportion, look for a number z such that

$$\frac{\text{part}}{\text{whole}} = \frac{60}{100} \text{ or } \frac{15}{z} = \frac{60}{100}$$

Hence, $z = \frac{(15)(100)}{60} = \frac{1500}{60} = 25$. That is, 15 is 60% of 25.

Percents Greater than 100%

Although the discussion about percent so far assumes a context of a part and a whole, it is not necessary that the part be less than the whole. In general, the whole is called the **base** of the percent. When the numerator of a percent is greater than the base, the percent is greater than 100%.

Example-13: 15 is 300% of 5, since

$$\frac{15}{5} = \frac{300}{100}$$

Example-14: 250% of 16 is 40, since

$$\left(\frac{250}{100}\right)(16) = (2.5)(16) = 40$$

Note that the decimal equivalent of 300% is 3.0 and the decimal equivalent of 250% is 2.5.